



Determination Of Constants of Elasticity

Determination Of Constants of Elasticity by Seares needle:-

Description:- Seares needle is a pair of exactly identical needles of either rectangular or circular cross-section; made of wood or metal. The needles have length about 20 cm and diameter 3 cm. Both the needle must also have same mass; so that they have same M.I about an axis passing through their centres and perpendicular to their lengths. The metal whose elastic constants are to be determined, is taken in the form of a short thin wire, of about 15 to 20 cm in length and 2 mm in diameter. This experimental wire is provided with two screws soldered at its two ends. & The Seares needles have holes at their centres, through which the two screws at the ends of the wire are passed and tightened by nuts, so that the needle and the wire form a T-combination.



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Theory: Given:

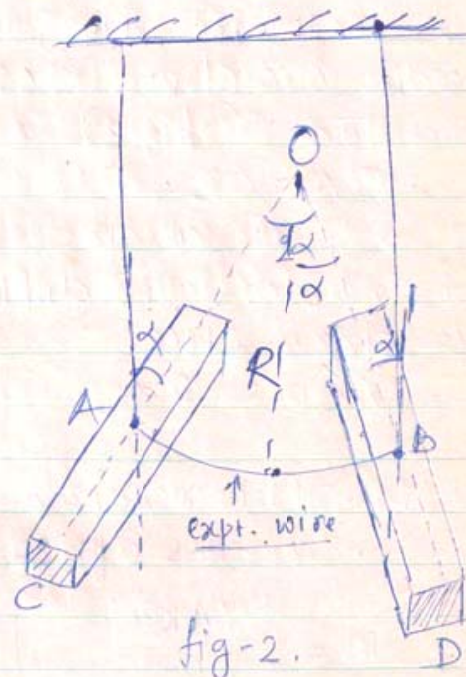
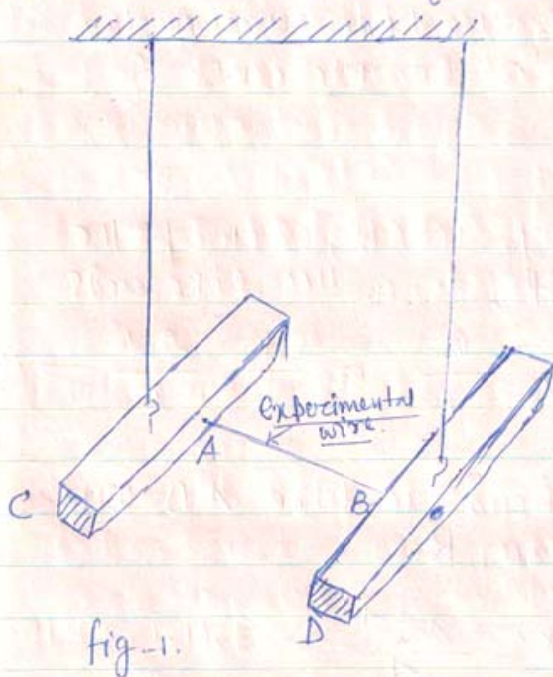
l = The length of the experimental wire AB.

r = Radius of the experimental wire AB.

Y & n = The Young's modulus and rigidity modulus of the experimental wire AB respectively.

I = M.I of each Searles needle, about an axis passing through the centre & perpendicular to the length of the needle.

(i) Determination of Young's modulus :- (Y):



The combination of ~~the~~ wire AB, fixed rigidly between the two needles C & D; is suspended by two threads of equal length from a fixed horizontal support (fig-1), so that H-combination lies on the horizontal plane. The like ends of the needle are drawn close, the wire AB is bent in the form of an arc and due to the flexural rigidity of the wire a restoring couple



equal and opposite to the bending moment is set up. When the needles are released, due to the restoring couple, the wire comes to its original position; the needles gain inertia & thus a torsional oscillation starts. Any one of the two needle is focussed by a telescope and the time of say 30 oscillations is measured by a stop watch and hence time period of oscillation can be calculated.

Theory: Let 'R' be the radius of curvature of the bent wire AB; when the like ends of the needles are drawn close & 'O' be the centre of curvature (fig-2)

$$OA = OB = R$$

let α = the angle through which, each of the two needles are deflected; when the like ends are drawn close.

$$\therefore \angle AOB = 2\alpha \quad \therefore 2\alpha = \frac{l}{R} \quad \text{or} \quad \frac{1}{R} = \frac{2\alpha}{l} \rightarrow (1)$$

let M = B.M required to bend the wire AB with radius of curvature R .

$$\therefore M = \frac{YAK^2}{R} \quad \left[AK^2 = \frac{\pi r^4}{4} = \text{G.M.I of the wire AB of circular cross-section} \right]$$

$$\text{or } M = YAK^2 \frac{2\alpha}{l} \quad \text{Putting Equation (1)}$$

$$\text{Restoring Couple} = C = -M = -YAK^2 \frac{2\alpha}{l} \rightarrow (2)$$

The negative sign indicates that the restoring couple is opposite in direction to the B.M. This restoring couple produces an angular acceleration in the needles.



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Using Newton's 2nd law :-

$$I \times \text{ang. accel}^n = C = -\frac{2\gamma AK^2}{l} \cdot \alpha$$

$$\text{or, ang. accel}^n = -\frac{2\gamma AK^2}{Il} \times \text{ang. displacement} \rightarrow (3)$$

For a given wire γ , AK^2 , l are constants & for a given needle I is constant from (3) :-

$$\text{ang. accel}^n \propto - \text{ang. displacement.}$$

Hence the torsional Oscillation of the needle is Simple harmonic.

let T_1 = Time period of the torsional Oscillation.

$$T_1 = 2\pi \sqrt{\frac{\text{ang. displacement}}{\text{ang. accel}^n}} \rightarrow (4)$$

Putting equation (3) in equation (4)

$$T_1 = 2\pi \sqrt{\frac{Il}{2\gamma AK^2}} \quad \text{Squaring both sides}$$

and rearranging :-

$$\gamma = \frac{4\pi^2 I \cdot l}{2AK^2 \cdot T_1^2}$$

$$\text{Putting } AK^2 = \frac{\pi r^4}{4}$$

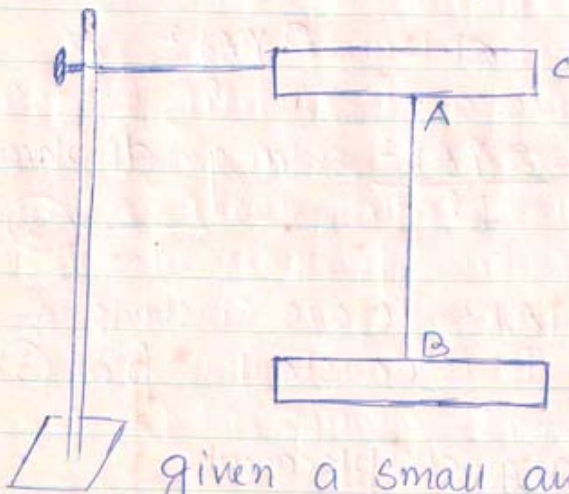
$$\gamma = \frac{4\pi^2 \cdot I \cdot l}{2 \cdot \frac{\pi r^4}{4} \cdot T_1^2}$$

$$\text{or } \boxed{\gamma = \frac{8\pi I l}{T_1^2 r^4}} \rightarrow (5)$$

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(ii) : To find the Rigidity modulus η :-



One of the two needles say 'C' is clamped on a stand, so that the wire AB hangs vertically carrying the other needle D at its free end. It then constitutes a torsional pendulum. The bob is

given a small angular displacement ' ϕ ' about the wire AB; the wire is twisted and due to the torsional rigidity, a restoring couple, equal and opposite to the torsional couple sets in which sets the bob in torsional oscillation. The time period of that torsional oscillation is measured.

$$C = \frac{\tau \propto \theta}{\theta} = \text{The torsional couple per unit twist.}$$

Torsional couple = $C\phi$. Restoring Couple = $-C\phi$
This restoring couple acting on the bob (needle) produces an angular acceleration. Using Newton's 2nd law :-

$$I \times \text{ang. accel}^2 = \text{Restoring couple} = -C\phi$$

$$\text{or } \text{ang. accel}^2 = -\frac{C}{I} \cdot \text{ang. displacement} \rightarrow (6)$$

Since for a given wire and a bob C & I are const.,
ang. accel² $\propto$ - ang. displacement.

Hence the torsional oscillation is simple harmonic.

Let T_2 = Time period of oscillation. $T_2 = 2\pi \sqrt{\frac{\text{ang. dis}}{\text{ang. acc}^2}}$

Putting Equation (6) in (7) :- $T_2 = 2\pi \sqrt{\frac{I}{C}}$

Continue



Squaring both sides and putting the value of c : Date _____

$$\frac{\pi n \delta^4}{2l} = \frac{4\pi^2 I}{T_2^2}$$

$$\text{or } \boxed{n = \frac{8\pi I l}{T_2^2 \delta^4}} \longrightarrow (8)$$

Using Equation (8) Rigidity modulus of the wire can be calculated.

Poisson's ratio σ :

from the relation between the elastic constants : $\gamma = 2n(1+\sigma)$

$$\text{or } \frac{\gamma}{2n} - 1 = \sigma$$

Putting Equations (5) and (8) :-

$$\sigma = \frac{8\pi I l / T_1^2 \delta^4}{2 \cdot 8\pi I l / T_2^2 \delta^4} - 1$$

$$\text{or } \boxed{\sigma = \frac{T_2^2 - T_1^2}{2 T_1^2}} \longrightarrow (9)$$

find K : $\gamma = 3K(1-2\sigma)$ Putting the values of γ and σ the value of K can be calculated.



Determination Of Constants of Elasticity

Let T_1 = Time period of Oscillation, when the solid cylinder occupy the outer positions (fig-1)

I_1 = The total M.I of the suspension in this case

T_2 = The time period of Oscillation, when the solid cylinders are kept in contact (fig-2)

I_2 = The M.I of suspension in this case.

Let D = length of the long cylinder.

$\frac{D}{4}$ = the length of each short cylinder = $2d$ (say)

m_1 = the mass of each solid cylinder.

m_2 = the mass of each hollow cylinder ($m_1 > m_2$)

Let $C = \frac{\pi n r^4}{2l}$, = Torsional Couple per unit twist of the suspension wire, the terms have their usual meaning.

Using the expression for time period of Torsional Oscillation:

$$T_1 = 2\pi \sqrt{\frac{I_1}{C}} \quad \text{or} \quad T_1^2 = \frac{4\pi^2 I_1}{C}$$

$$T_2 = 2\pi \sqrt{\frac{I_2}{C}} \quad \text{or} \quad T_2^2 = \frac{4\pi^2 I_2}{C}$$

$$\text{or } T_1^2 - T_2^2 = \frac{4\pi^2}{C} (I_1 - I_2)$$

$$\text{or } C = \frac{\pi n r^4}{2l} = \frac{4\pi^2 (I_1 - I_2)}{T_1^2 - T_2^2}$$

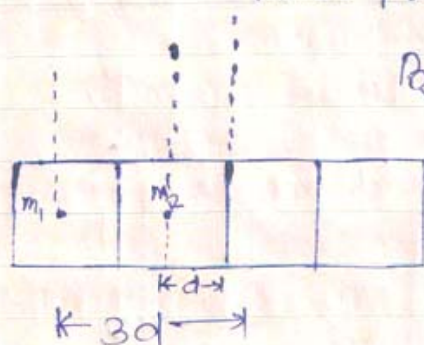
$$\text{or } n = \frac{8\pi l (I_1 - I_2)}{r^4 (T_1^2 - T_2^2)} \longrightarrow \textcircled{1}$$



To find $I_1 - I_2$:

Let $I_s = M.I$ of each solid cylinder about the axis of oscillation.

$I_h = M.I$ of the each hollow cylinder about the axis of oscillation when it occupies the inner position.



By applying theorem of 11^{th} axis.

$$I_s = I_{cs} + m_1 (3d)^2$$

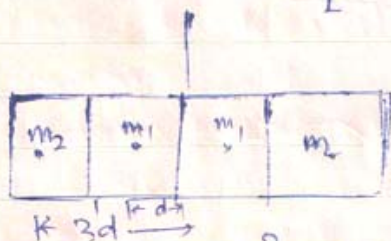
$$I_h = I_{ch} + m_2 (d)^2$$

The total M.I of the suspension about the axis of oscillation.

$$I_1 = I_0 + 2 [I_s + I_h]$$

$I_0 = M.I$ of the long needle about the axis of oscillation.

$$I_1 = I_0 + 2 [I_{cs} + I_{ch} + 9m_1 d^2 + m_2 d^2] \quad \text{--- (2)}$$



Similarly following 2^{nd} (2) :-

$$I_2 = I_0 + 2 [I_{cs} + I_{ch} + 9m_2 d^2 + m_1 d^2] \quad \text{--- (3)}$$

Subtracting Equation (3) from (2) :- $(2) - (3) \Rightarrow$

$$I_1 - I_2 = 8d^2 (m_1 - m_2)$$

$$\text{or, } I_1 - I_2 = 8 \cdot \frac{D^2}{64} (m_1 - m_2) \quad \text{--- (4)}$$

Putting (4) in (1) :-
$$n = \frac{8\pi l (m_1 - m_2) D^2 / 8}{8^4 (\tau_1^2 - \tau_2^2)}$$



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$$\eta = \frac{\pi D^2 L (m_1 - m_2)}{8^4 (T_1^2 - T_2^2)} \longrightarrow (5)$$

This method is very accurate for determination of Rigidity modulus.